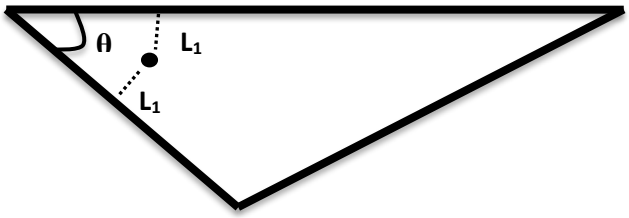
WT SOFTWARE-BASED LABORATORY

Session 6: Individual Assignment

In this exercise you are required to analyse the drawdown and build-up well test data (pressure versus time) for a well located in a reservoir compartment. It is believed that the drainage area of this well can be approximated by a rhombus as shown in figure below. The well, reservoir and production histories are as follows: Surface production rate=6000 STB/d, viscosity=3 cp, Porosity =0.25, Oil formation factor 1.25 bbl/STB, Total compressibility=3e-5 psi-1, Effective producing interval =30 ft, wellbore radius =0.3 ft, production period=3000 hrs. The recorded pressure history (pressure vs time) is included in the file "2025PanSystemDataEx6FinalA.dat", which accompanies this document.



Load the data into PanSystem, analyse the drawdown and build-up period and perform the tasks requested below.

Drawdown (DD) analysis

1. Show the ETR, MTR and LTR regions and the flow regimes within each region on your constructed diagnostic derivative log-log plot. For each flow regime, identify the key signatures. (13 marks)

2. (A) Prepare two separate specialised plots that display the wellbore storage coefficient, average permeability and the skin factor values obtained by PanSystem.
(B) Then manually calculate wellbore storage coefficient, average permeability and the skin factor. (13 marks)

3. Use the type-curve matching facility in PanSystem to obtain the parameters mentioned in step two (show the corresponding plot). (5 marks)

4. There are different reasons for early time not following the middle time semi-log straight line behaviour.
(A-i) Discuss three reasons; two mathematically based and one due to operational issue that can cause such ETR deviations and (A-ii) highlight how their signatures can be distinguished on a draw-down semi-log plot.
(B) Discuss (i) which one is more likely to affect the results (ii) and how to deal with such a deviation if the data have already been affected by it. (14 marks)

5. (A) Produce a specialised plot which shows the distance to the fault obtained by PanSystem.
(B) Then using slopes and intercepts of this plot, manually calculate the rhombus angle θ and the distance to the two nearest boundaries. (9 marks)

6. (A) Prepare the specialised plot that shows the well shape factor and drainage area values obtained by PanSystem.
(B) Then manually calculate the well shape factor and drainage area values.
(9 marks)

7. Quick Match and Auto-match Non-linear Regression can help in analysing well test data. Compare these two methods with Straight Line Analysis highlighting when they should be used and for what purposes. (10 marks)

Build-up (BU) analysis

8. Prepare the specialised plots that show the wellbore storage coefficient, average permeability and skin factor values from BU analysis. (6 marks)

9. Comment on the two approaches allowing us to use DD type curves for analysing BU data if the preceding DD was IA (include one advantage and one disadvantage for each). (12 marks)

10. Manually calculate the average reservoir pressure from the MBH method. (9 marks)
(Hint: You may use the suitable equation for $(P_D)_{MBH}$ mentioned in your course notes.)

*Note

- Your submission should not exceed 9 pages (text font: Times 12 with 1.5 line spacing or equivalent) excluding the front sheet (please do not include the question sheet).
- The report should have a cover page with your full name and student number and be submitted on Canvas.
- For your manual calculations, include all steps including substitution of variables. No marks can be given for an incorrect answer if the method of calculation is not presented, and no complete mark will be awarded to a correct final answer if intermediate steps are not included.
- The answer to all questions especially descriptive part of any questions should be your own otherwise you will be penalised.
- These ensure fair and clear marking.

NAYEBALE VALENTINE

HOO38OO49

G11WT RESERVOIR ENGINEERING WELL TEST ANALYSIS COURSE WORK

2025.

DRAWDOWN (DD) ANALYSIS

QUESTION ONE.

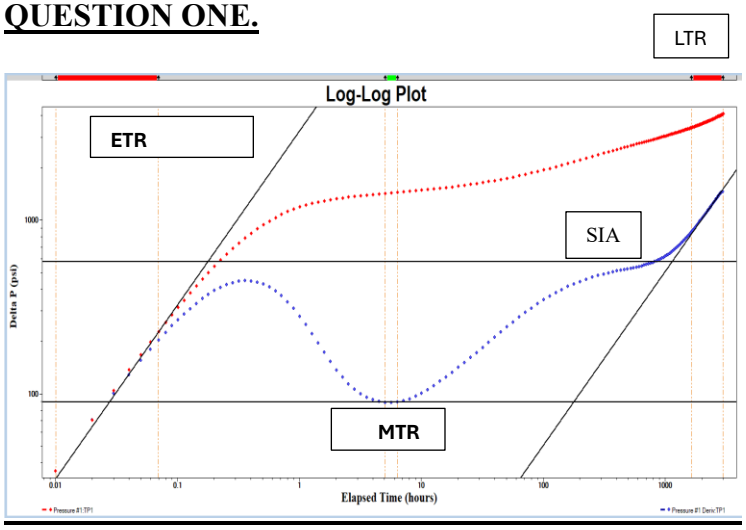


FIGURE 1

Early time region (ETR): In the early times of production from the reservoir (ETR), wellbore storage and skin dominate the flow of fluids. This is identified by the unit slope on the pressure derivative curve on the log-log diagnostic plot. The point where the derivative curve starts to shift away from the change in pressure curve is where the wellbore storage effect ends. The transition from early time region to middle time region (MTR) is identified by the hump like shape of the derivative curve. This is due to an increase in the pressure drop caused by skin around the wellbore. The presence of skin restricts the flow of fluids thus causing an increase in the pressure drop.

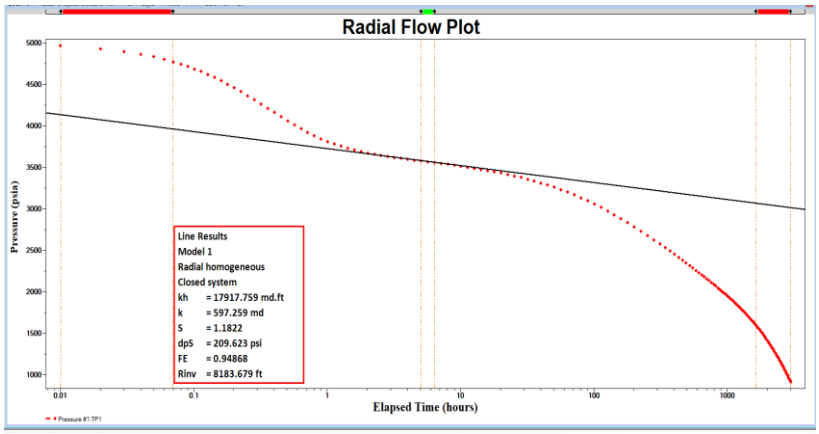
Middle time region (MTR): This period is identified by the zero-slope line indicating radial flow regime. Once the derivative pressure reaches a peak in the early time region, gradually declines until it stabilises into a horizontal plateau. This is termed as the zero slope line and thus defines the middle time region. Here the reservoir is said to be infinitely acting because the pressure wave is moving radially across the reservoir and hasn't reached the boundary. The flow of the fluids is now a function of the reservoir parameters such as permeability, skin or formation transmissibility.

Semi-infinite acting. (SIA): After the MTR, there is a transition to the semi-infinite acting flow regime. This is shown by the second horizontal slope formed by the derivative curve. This is because the pressure wave has met the first boundary that doesn't flow. This indicates the two intersecting faults that do not flow as observed in the reservoir model.

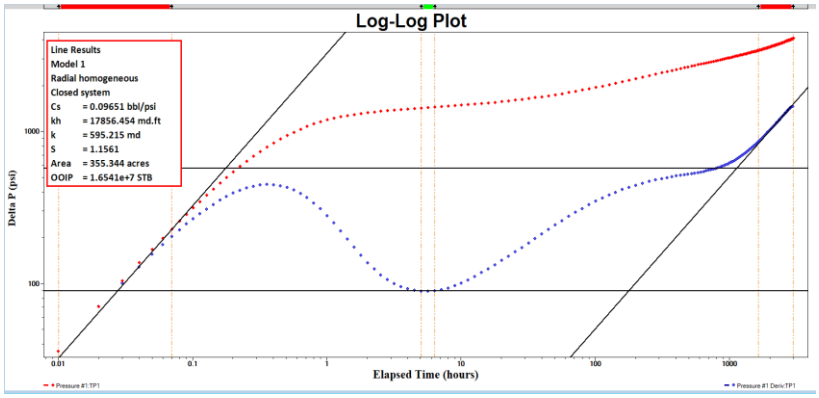
Late time region (LTR): This region is identified by a unit slope on the derivative curve as it rises at the end of the semi-infinite acting flow regime. The shows that pressure wave has reached the final boundary and fluid flow is dominated by the boundary effects. The system is said to have reached the semi-steady state flow regime, where the pressure change due to the flow of fluids is constant with time.

QUESTION TWO

A) SEMI-LOG SPECIALISED PLOT



LOG-LOG DIAGONISTIC PLOT



B) wellbore coefficient calculation.

Line Results

Line Creation Type :	Unit slope
Slope :	1
Intercept :	3238.12045455741
Coefficient of Determination :	Not Used
Line Type :	Wellbore storage

$$C_s = \frac{qs\ x\ Bo}{24\ x\ dPint}$$

$$C_s = \frac{6000\ x\ 1.25}{24\ x\ 3238.12}$$

C_s = 0.096506615

Average permeability using log-log equation.

$$K = \left[\frac{141.2qsBo\mu}{h} x \frac{0.5}{dp}\right] \quad K = \frac{141.2\ X\ 6000\ X\ 1.25\ X\ 3}{30}\ X\ \frac{0.5}{89.9594} \quad K = \mathbf{588.5988568md.}$$

Average permeability of semi-log equation

$$K = \frac{162.2qsBo\mu}{mh}$$

$$K = \frac{162.2\ X\ 6000\ X\ 1.25\ X\ 3}{204.136\ X\ 30} \qquad K = \underline{\mathbf{595.9262452md}}$$

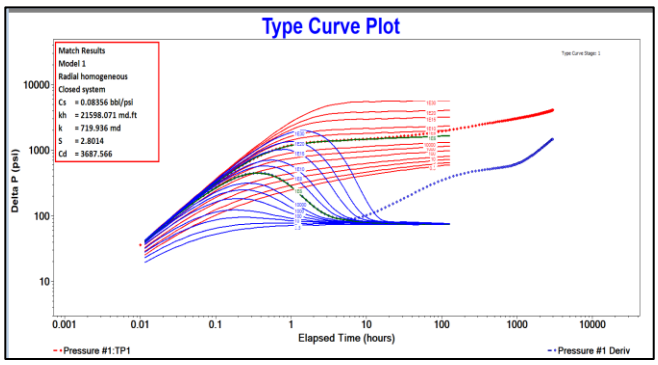
Skin factor calculation using log-log equation.

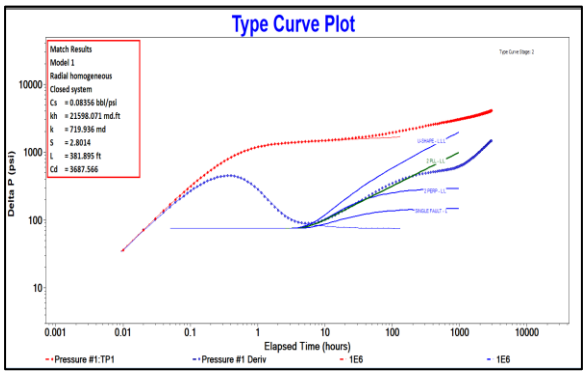
$$S = 1.1513\left[\frac{Pi-P1hr}{m} - \log_{10}\left[\frac{kdt}{\Phi\mu Ctrw}\right]\right. + 3.2275$$

$$S = 1.1513\left[\frac{5000-3720.261108}{204.136} - \log_{10}\frac{595.9262452}{0.25\ x\ 3\ x\ 3e{-}5\ x\ 0.09}\right] + 3.2275$$

<u>Skin factor = 1.183287</u>.

<u>QUESTION 3</u>





<u>QUESTION 4</u>

ai) mathematical reason.

Wellbore storage: during the early time production, the flow of fluids is dominated by the wellbore storage which shadows the true pressure response of the reservoir as fluids flow. Therefore, the production of the fluids does not represent the flow of the fluids in the wellbore. This causes a deviation of the straight line as observed during the middle time region where fluids flow radially, making it difficult to estimate the reservoir parameters such as permeability that influence fluid flow.

Skin: skin causes restriction of fluids in the wellbore causing an additional pressure to drop near the wellbore. The additional pressure drop on the reservoir caused by positive skin would result into a lower production rate however negative skin improves production due to the stimulation of the well.

aii)

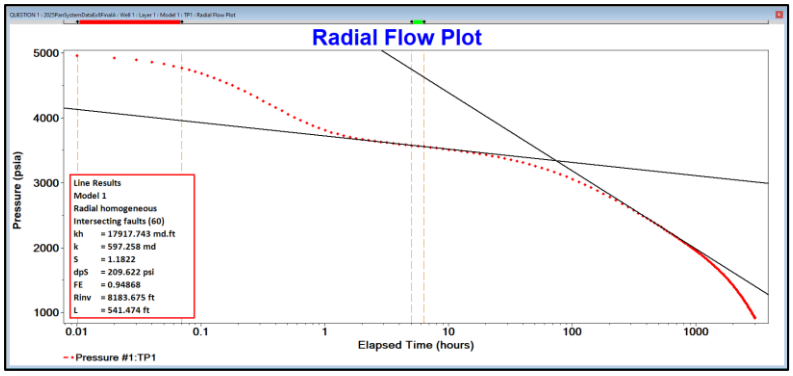
- wellbore storage forms a smooth curve of pressure points in the early time region of which later, the pressure drop stabilizes to form a straight line in the middle time region.
- Positive skin results in higher pressure drop forming a steep vertical line while a negative skin results in low pressure drop forming a gently sloping straight line on the semi-log plot.
- On the semi-log plot, changing the flow rate would result in an inconsistent change in pressure transient. This would manifest into scattering of pressure points that deviate away from the straight line that should be observed.

bi) wellbore storage is most likely to affect the pressure transient results in the early time region. Wellbore storage dominates the flow rate in the early time region. The pressure response due to fluid flow in the wellbore is controlled by compressibility of the fluids and the wellbore conditions, thus shadowing the true pressure transient behaviour of the reservoir due to radial flow of fluids. Since the pressure response is dominated by wellbore storage, it masks the true behaviour of the reservoir therefore the reservoir model type cannot be analysed.

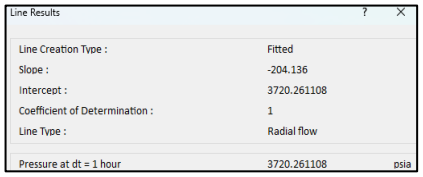
bii) type curve matching would enable the accurate analysis of the pressure transient by accounting for the wellbore storage in the early time region. By finding the right type-curve for the real time pressure data that is deviating away from the MTR behaviour, it separates the effects of wellbore storage that shadow the true pressure response of the reservoir. This gives a clear estimate of the permeability, skin, drainage area and shape factor.

QUESTION 5

5A)



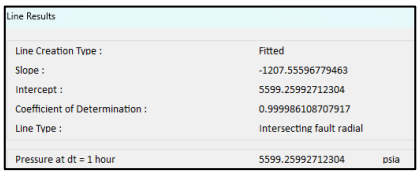
5B)



Line Results

Line Creation Type :	Fitted
Slope :	-204.136
Intercept :	3720.261108
Coefficient of Determination :	1
Line Type :	Radial flow
Pressure at dt = 1 hour	3720.261108 psia

LINE DETAILS FOR SLOPE OF MTR



Line Results

Line Creation Type :	Fitted
Slope :	-1207.55596779463
Intercept :	5599.25992712304
Coefficient of Determination :	0.999986108707917
Line Type :	Intersecting fault radial
Pressure at dt = 1 hour	5599.25992712304 psia

LINE DETAILS OF SLOPE OF LTR

Calculation for rhombus angle.

Slope ratio = $\frac{slope\ of\ LTR}{slope\ of\ MTR}$ slope ratio = $\frac{-1207.55596779463}{-204.136}$ slope ratio = **5.915448** $\theta = \frac{360}{5.915448}$

θ = **60.85°**

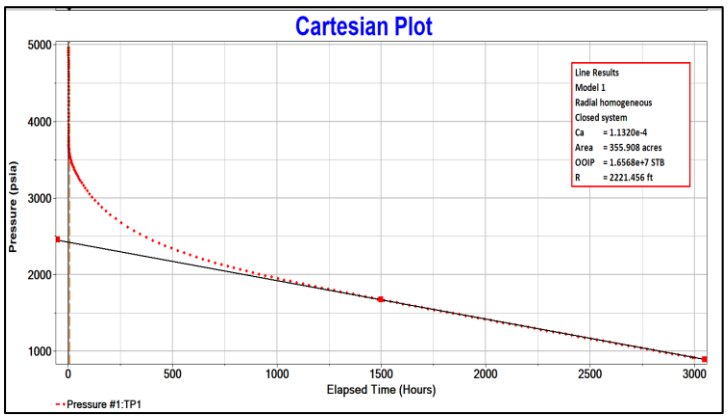
calculation for distance from the fault.

L = 0.01217 $\sqrt{\frac{k\Delta tex}{\Phi\mu Ct}}$ L= 0.01217 $\sqrt{\frac{597.258\ x\ 74.767}{0.25\ x\ 3\ x\ 3e-5}}$ L= 0.01217 $\sqrt{\frac{44655.18889}{2.25E-05}}$

L = 542.170 ft **distance from the fault** = 542.170 ft

QUESTION 6.

6A)

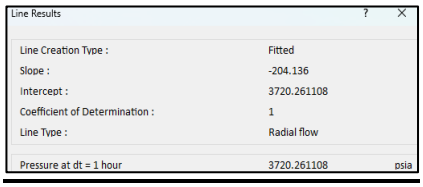


6B)

Drainage area calculation

A = - $\frac{0.23395qsBo}{43560\Phi Cthm*}$ A= $\frac{0.23395\ x\ 6000\ x\ 1.25}{43560\ x\ 0.25\ x\ 3e{-}5\ x\ 30\ x{-}0.5013}$ **A = 355.907635 acres.**

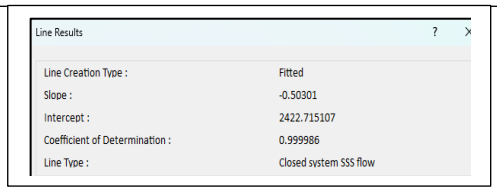
Shape factor calculation.



Line Results

Line Creation Type :	Fitted
Slope :	-204.136
Intercept :	3720.261108
Coefficient of Determination :	1
Line Type :	Radial flow
Pressure at dt = 1 hour	3720.261108 psia

line details for slope on semi-log plot



Line Results

Line Creation Type :	Fitted
Slope :	-0.50301
Intercept :	2422.715107
Coefficient of Determination :	0.999986
Line Type :	Closed system SSS flow

line details for slope of line on cartesian plot

$$C_A = 5.456\ \frac{|m|}{|m*|}\ e^{-2.303(P1h-P*int)/|m|}$$

$$C_A = 5.456\ \frac{|-204.136|}{|-0.50301|}\ e^{-2.303(3720.261108-2422.715107)/|-204.136|}$$

C_A = 9.7223e-04.

Question 7.

Quick match simulation can be used to account for the wellbore storage effect that affect the pressure transient in the early period that mask the reservoir performance and deviate away from the straight line observed in MTR. By fitting the right theoretical curve to the real-time data of the reservoir model. The reservoir properties (permeability and skin) can be adjusted to fit the right type-curve which would help in understanding the reservoir characteristics. This method enables for accurate calculation of the permeability, skin, drainage area of the reservoir by relating the respective reservoir model to the type-curve that shows the characteristic of wellbore storage in ETR. This method is mostly used when the behaviour of pressure transient in the early time does not behave the same as the middle time region and deviates away from the straight line.

Auto-match Non-linear Regression is a software algorithm that matches a mathematical theoretical type-curve to the reservoir model and try and accurately determine the reservoir properties. This method is used for very heterogeneous reservoirs which have un-uniform reservoir properties, non-radial flow and complex geometry. This method uses high precision

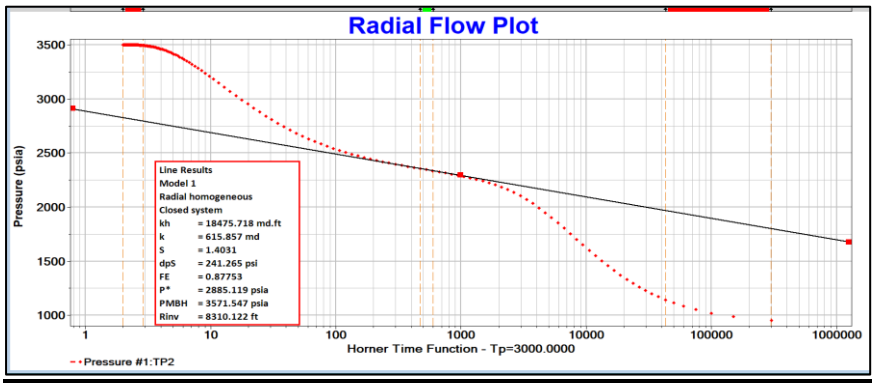
to find the best match of the type curves that fits the respective reservoir model, and with this can accurately analyse the behaviour of the reservoir as fluids flow. The limitation with this method is that if a wrong model is chosen, then it will result into a wrong characteristic analysis of the reservoir and wrong calculations for permeability and skin

Straight line analysis is done by fitting a best line fit of a straight line within the points plotted on a semi-log plot to analyse radial flow of fluids. If the pressure drop is constant with a log of time, then a straight line is observed, which represents the radial flow of fluids, especially in the middle time region. The slope of the straight line is used to determine the permeability of the reservoir, which can help to analyse the behaviour of the reservoir's fluid flow. The pressure intercept P(int) of the slope is used to estimate the skin, which restricts the flow of fluids. Deviations of the straight line in the late time region would indicate the type of reservoir boundary. If the slope of the line doubles, then it could indicate the first no flow such as a fault, if the pressure drop stabilises to zero slope, then it could indicate the final reservoir boundary. This method is used to analyse reservoirs that are homogenous and have uniform properties across their geological geometry.

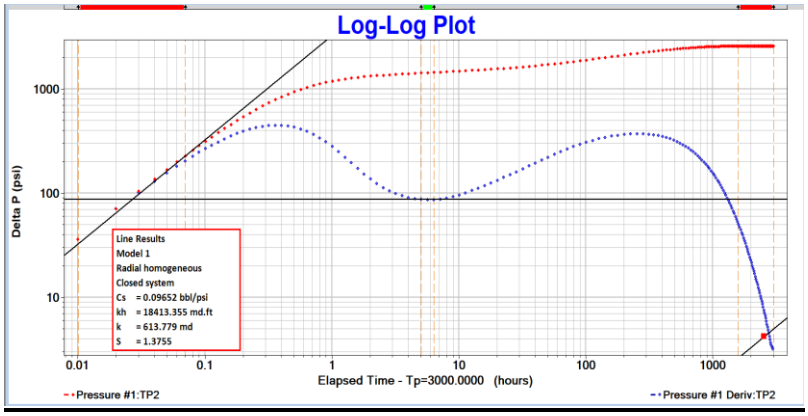
BUILD-UP (BU) ANALYSIS

Question 8

SEMI LOG PLOT.



LOG-LOG PLOT



QUESTION 9

The superposition principle using the Horner's method approach helps to analyse the pressure transient during build up by relating it with the drawdown type curve plot. The pressure response during buildup time comes from the total pressure recovery during buildup time and the pressure drop due to the influence of drawdown before shut in. Horner's method adjusts the time to account for the pressure drop still present immediately after shit in period, allowing us to find the relationship between the total build up pressure and Horner's time on a semi-log plot.

Advantage: The Horner's plot obtained can easily be compared with the drawdown type curves, by finding one that has a familiar shape to it. This would give insights on the reservoir model, enabling us to calculate the permeability and skin factor.

Disadvantage: It is only accurate if the reservoir pressure was infinitely acting before the shut-in period and doesn't account for the pressure drop due to the boundary effects.

The Agarwal method removes the effect of the pressure drop immediately after shut in period and only accounts for he times the pressure was recovered during the build-up time. Therefore, the equivalent Agarwal time and the stabilized pressure is anlaysed to understand the reservoir characteristics.

Advantage: The Agarwal equivalent time enables the accurate analysis of the pressure build up about the drawdown type types, because the plot of the pressure builds up across that time assumes the behaviour of the drawdown test making it easier and more correctly to estimate permeability and skin.

Disadvantage: If the production period was short, the infinitely acting period was short-lived and the flow of fluids was mostly dominated by wellbore storage, then the Agarwal method

would not be accurate in determining the beginning period of the pressure build up after the shut-in period regardless of the time.

QUESTION 10.

Dimensionless time calculated drainage area.

$t_{DA} = \frac{0.00026367Ktp}{\Phi\mu CtA43560} = t_{DA} = \frac{0.00026367tp}{0.25*3*3e-5*355.907635*43560}$ t_{DA} = **1.39654478477.**

when tp>tsss, then semi- steady state has been reached for drawdown. when we substitute into the MBH equation. $P_{D,MBH} = \ln(C_A * t_{Pda})$ $P_{D,MBH} = \ln(9.7223E\text{-}04 \times 1.39654478477)$ $P_{D,MBH}$ = **-6.60191401**

calculating for the P_{av}.

$P_{D,MBH} = 2.303 * \frac{(P* - P\ av)}{|m|}$

$-6.60191401 = 2.303 * \frac{(2885.19 - Pav)}{|197.9708|}$

$P_{av.} = 2885.19 - \frac{(-6.60191401 \times |197.9708|)}{2.303}$

P_{av} = 3452.633 psi.